

ERASTEEL TECHNICAL DATASHEET: ASP APZ10

Application Family: Specialties | Standards: Martensitic Stainless PM Tool Steel

1. Grade Overview & Characteristics

ASP APZ10 is a stainless martensitic powder metallurgy steel with 19% Chromium. Combines abrasive wear resistance with high corrosion resistance in acidic or chemically aggressive environments.

2. Nominal Chemical Composition (wt. % range)

Element	Carbon (C)	Chromium (Cr)	Molybdenum (Mo)	Tungsten (W)	Vanadium (V)	Cobalt (Co)	Other
Range %	1.20 - 1.30	18.5 - 19.5	2.0 - 2.2	0.0	0.7 - 0.9	0.0 0.1%	Nitrogen for corrosion resistance

3. Mechanical Hardness & Property Profiles (Tempered state)

Metric	Value	Metric	Value (Rating 1-10)
Working Hardness Range	58.0 - 60.0 HRC	Machinability	6.0/10
Max Reachable Hardness	60.0 HRC	Wear Resistance	7.0/10
Annealed Delivery State	Max 280-350 HB	Toughness	7.0/10
Hot Hardness Retention	Secondary hardening	Hot Hardness	4.0/10
Heat Treatment Type	Vacuum / Salt bath	Grindability	7.0/10

4. Industrial Applications

Plastic injection molds for corrosive polymers (PVC, fluoropolymers), food processing knives, medical instruments, valve components.

5. Heat Treatment & Tempering Guidelines

Austenitizing at 1050-1080°C. Deep freezing recommended (-70°C). Double tempering at 200-250°C (corrosion resistant) or 500-520°C (secondary hardening). Hardness 58-60 HRC.